

Synopsis

OfflineBridge : A Progressive Web App for Rural Digital Services

Department: *Information Science & Engineering*

The National Institute of Engineering, Mysuru — Academic Year: 2026–27

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Project Title: “OfflineBridge : A Progressive Web App for Rural Digital Services.”

Abstract:

The project aims to develop digital services are increasingly used for government and rural public services such as agriculture, scholarships, certificates, pensions, welfare schemes and grievance registration. However, rural users may face unstable internet connectivity, limited digital literacy and difficulty accessing online services. In conventional web applications, a network failure during form filling or submission may interrupt the process and cause loss of entered information.

To address these problems, OfflineBridge is proposed as an **offline Progressive Web Application (PWA)** for rural digital services. The application allows users to access selected services and fill forms even when the internet is temporarily unavailable. The entered data is stored locally using **IndexedDB** and automatically synchronized with the backend server when the internet connection is restored. The system also provides low-data operation, scheme eligibility assistance, application tracking and grievance services. The project aims to make digital services more reliable, accessible and easier to use for rural communities.

Objectives:

1. Offline Digital Access: Develop a PWA that allows rural users to access and use selected digital services even without continuous internet connectivity.

2. Offline Data Storage & Synchronization: Store application data locally using IndexedDB and automatically synchronize it with the server when connectivity is restored.

- 3. Reliable Service Access:** Reduce data loss, repeated form filling and interruptions caused by network failures.
- 4. Low-Data Operation:** Enable the application to work efficiently under weak or limited network conditions.
- 5. Government Service Assistance:** Provide information about selected government services and a simple scheme eligibility assistant.
- 6. Application & Grievance Tracking:** Allow users to track application status and submit or monitor grievances.
- 7. Simple User Experience:** Develop a simple, accessible and user-friendly interface for rural citizens.

Software Requirements:

Operating System	: Windows 10/11, Linux (Ubuntu)
Development Platform	: VS Code
Frontend Framework	: React.js
Web Technologies	: HTML, CSS, JavaScript
Application Type	: Progressive Web Application (PWA)
Offline Support	: Service Worker
Local Database	: IndexedDB
Backend Environment	: Node.js
Backend Framework	: Express.js
Central Database	: MySQL
Web Browser	: Google Chrome / Microsoft Edge
Version Control	: Git & GitHub

Hardware Requirements:

Processor	: Intel Core i3/i5 or equivalent
Memory	: Minimum 8 GB RAM
Storage	: Minimum 10GB.

Guide:

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